

Emanuele Nardone, PhD

Postdoctoral Researcher — Machine Learning and Pattern Recognition

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Italian citizen | Currently based in Joensuu (FI) | Italian (native), English (C1) | Updated 09/2026

PROFILE

PhD in Artificial Intelligence (10/2025) working on machine learning and pattern recognition methods for decision support: **evolutionary computation and feature selection, multimodal classification and classifier combination, uncertainty quantification via conformal prediction, and explainable AI**. The methodological core of my work concerns handwriting analysis for the early detection of cognitive and developmental disorders, published as first author in *Engineering Applications of Artificial Intelligence*, *Computers in Biology and Medicine* and *Pattern Recognition Letters*, and recognised by the **Best EvoApps Paper Award at EvoStar 2026**. Since 02/2026 I am a postdoctoral researcher at the University of Eastern Finland, where I transfer the same methods to inverse problems, spectral data analysis and image segmentation in photonics (Horizon Europe **HERMES**, AI/ML lead). Teaching record as **adjunct lecturer in Artificial Intelligence** (MSc LM-32) and course tutor, with **62 h** of documented university teaching, and co-supervision of an **Erasmus Mundus Joint Master** research internship. **Guest Editor** of a special issue of *Big Data and Cognitive Computing*, Workshop Chair (BIOMAP@ICPR 2026) and reviewer for eight international journals.

EDUCATION

- **PhD in Artificial Intelligence** — University of Cassino and Southern Lazio (IT). **11/2022 – 10/2025**
Dept. of Electrical and Information Engineering (DIEI). Thesis: *AI-Based Handwriting Analysis for Early Detection of Developmental and Cognitive Disorders*. Supervisors: Prof. F. Fontanella, Prof. C. De Stefano; co-supervisor: Prof. L. Vanneschi.
- **Visiting PhD Researcher** — NOVA IMS, Universidade Nova de Lisboa (PT). **05/2024 – 08/2024**
Genetic Programming for multimodal classification (host: Prof. L. Vanneschi).
- **MSc in Software Engineering** (LM-32) — University of Cassino and Southern Lazio. **03/2021**
Final mark **110/110 *summa cum laude***. Erasmus+ at FEUP, University of Porto (PT), 2020.
- **Esame di Stato** — professional qualification to practise engineering (IT).

ACADEMIC AND RESEARCH POSITIONS

- **Postdoctoral Researcher** — University of Eastern Finland, Joensuu (FI). **02/2026 – present**
Center for Photonics Sciences (PROFI6 programme). ML/DL for optical and spectral data, inverse-problem reconstruction, image segmentation; AI/ML lead on HERMES (Horizon Europe). Supervisors: Prof. P. Kuzhir, Prof. G. Fedorov.
- **Industrial Secondment** — Monozukuri S.p.A., Rome (IT), UEF-funded under HERMES. **2026, 4 months**
Feasibility study on THz-based non-invasive glucose detection; design of AI/ML signal-processing pipelines for THz spectroscopic biosensing data.
- **Research Fellow** — University of Cassino and Southern Lazio, DIEI. **04/2021 – 08/2022**
Two consecutive research fellowships. ML pipelines for the early diagnosis of cognitive impairment from handwriting; design and development of C#/WPF acquisition software for Wacom digitising tablets.

SELECTED PUBLICATIONS

Twelve peer-reviewed publications selected from a total of 18 documents indexed in Scopus. Preprints and non-refereed material are excluded. The candidate is **first author in 7 of the 12** works listed; candidate's name in bold. Within each category, first-authored works are listed first.

Articles in international peer-reviewed journals

- P1. Nardone E.**, D'Alessandro T., De Stefano C., Scotto di Freca A., Fontanella F. (2026). *Artificial intelligence for Alzheimer's disease screening via handwriting images acquired by smartphones*. **Engineering Applications of Artificial Intelligence**, 182, 116023. ISSN 0952-1976. doi:10.1016/j.engappai.2026.116023.
- P2. Nardone E.**, De Stefano C., Cilia N.D., Fontanella F. (2025). *Handwriting strokes as biomarkers for Alzheimer's disease prediction: a novel machine learning approach*. **Computers in Biology and Medicine**, 190, 110039. ISSN 0010-4825. doi:10.1016/j.compbimed.2025.110039.
- P3. Nardone E.**, D'Alessandro T., De Stefano C., Fontanella F., Scotto di Freca A. (2025). *A Bayesian network combiner for multimodal handwriting analysis in Alzheimer's disease detection*. **Pattern Recognition Letters**, 190, 177–184. doi:10.1016/j.patrec.2025.02.019.
- P4. Nardone E.**, D'Alessandro T., De Stefano C., Fontanella F. (2025). *How data augmentation affects evolutionary algorithms in feature selection: an experimental study*. **SN Computer Science**, 6, 536. ISSN 2661-8907.

doi:10.1007/s42979-025-04049-3.

- P5.** Fontanella F., D'Alessandro T., **Nardone E.**, De Stefano C., Vicidomini C., Roviello G.N. (2026). *Artificial intelligence for natural products drug discovery in neurodegenerative therapies: a review*. **Biomolecules**, 16(1), 129. ISSN 2218-273X. doi:10.3390/biom16010129.

Papers in peer-reviewed international conference proceedings

- P6.** **Nardone E.**, De Stefano C., Scotto di Freca A., Fontanella F., D'Alessandro T. (2026). *Toward reliable uncertainty quantification in surrogate-assisted evolutionary algorithms via temporal conformal prediction*. **EvoApplications 2026** (EvoStar), *Applications of Evolutionary Computation*, Lecture Notes in Computer Science, Springer, pp. 384–401. ISBN 978-3-032-23604-3. **Best EvoApps Paper Award**. doi:10.1007/978-3-032-23604-3_24.
- P7.** **Nardone E.**, D'Alessandro T., De Stefano C., Fontanella F. (2025). *Evolutionary computation for causality-driven feature selection: a preliminary study*. **EvoApplications 2025** (EvoStar), *Applications of Evolutionary Computation*, Lecture Notes in Computer Science, Springer, pp. 385–401. ISBN 978-3-031-90062-4. doi:10.1007/978-3-031-90062-4_24.
- P8.** **Nardone E.**, Cantone M., Lozupone G., Pace C.D., Russo C., D'Alessandro T. (2026). *Scale-aware fractal-lacunarity analysis of handwriting for Alzheimer's disease classification*. **ICIAP 2025 International Workshops, Image Analysis and Processing**, Lecture Notes in Computer Science, Springer, pp. 41–53. ISBN 978-3-032-11381-8. doi:10.1007/978-3-032-11381-8_4.
- P9.** Lozupone G., **Nardone E.**, Pace C.D., D'Alessandro T. (2024). *Transformers and CNNs in neurodiagnostics: handwriting analysis for Alzheimer's diagnosis*. **ICPR 2024** — 27th Int. Conf. on Pattern Recognition, *Pattern Recognition*, Lecture Notes in Computer Science, Springer, pp. 447–463. ISBN 978-3-031-78195-7. doi:10.1007/978-3-031-78195-7_30.
- P10.** D'Alessandro T., De Stefano C., Fontanella F., **Nardone E.**, Pace C.D. (2024). *From handwriting analysis to Alzheimer's disease prediction: an experimental comparison of classifier combination methods*. **ICDAR 2024** — 18th Int. Conf. on Document Analysis and Recognition, Lecture Notes in Computer Science, Springer, pp. 334–351. ISBN 978-3-031-70536-6. doi:10.1007/978-3-031-70536-6_20.
- P11.** D'Alessandro T., De Stefano C., Fontanella F., **Nardone E.** (2024). *Integrating data augmentation in evolutionary algorithms for feature selection: a preliminary study*. **EvoApplications 2024** (EvoStar), *Applications of Evolutionary Computation*, Lecture Notes in Computer Science 14634, Springer, pp. 397–412. ISBN 978-3-031-56852-7. doi:10.1007/978-3-031-56852-7_25.
- P12.** D'Alessandro T., De Stefano C., Fontanella F., **Nardone E.**, Scotto di Freca A. (2024). *A genetic algorithm for feature selection for Alzheimer's disease detection using a deep transfer learning approach*. **WIVACE 2023** — Italian Workshop on Artificial Life and Evolutionary Computation, Communications in Computer and Information Science 1977, Springer, pp. 309–323. ISBN 978-3-031-57430-6. doi:10.1007/978-3-031-57430-6_24.

Bibliometric indicators Scopus (as of 09/2026): **18** documents, **48** citations from 41 citing documents, **h-index 5**. Author ID 58599513400.

TEACHING ACTIVITY

Period	Role, course and programme	Hours
A.Y. 2025/26	Adjunct Lecturer (docente a contratto) — <i>Artificial Intelligence</i> , MSc in Computer Engineering (LM-32), University of Cassino and Southern Lazio. Python for ML, supervised learning, metaheuristic optimisation.	14
A.Y. 2024/25	Adjunct Lecturer (docente a contratto) — <i>Artificial Intelligence</i> , MSc in Computer Engineering (LM-32), University of Cassino and Southern Lazio.	14
02–05/2025	Tutor — Advanced Training Course on <i>Generative AI</i> , University of Cassino. Hands-on notebooks, ethics and reproducibility; supervision of participants' final projects.	30
01/2025	Invited seminar — <i>Kubernetes and GPU resource management</i> , AIDA LAB, University of Cassino.	2
07/2024	Invited seminar — <i>Frontiers in AI: generative models, self-supervised learning, diffusion</i> , University of Cassino.	2
Total university teaching		62
01/2025	Tutor, PCTO programme — introduction to ML and Python for secondary-school students, University of Cassino (pre-university outreach).	20

Supervision and teaching credentials

- **Co-supervisor, MSc research internship** — *iPSRS* Erasmus Mundus Joint Master (Intelligent Photonics for Security, Reliability, Sustainability & Safety). **06–08/2026**

Student: Mahamudur Raman. Host: Dept. of Physics and Mathematics, University of Eastern Finland. Co-supervisor: A. Saushin. Thesis: *Development of a Raman hyperspectral preprocessing and spectral ratio analysis pipeline for the THz-Skin project*.

- **IEEE Trainer for Europe** (11/2024) — cross-European trainer credential (IEEE Region 8).

PARTICIPATION IN FUNDED RESEARCH PROJECTS

- **HERMES** — Horizon Europe, UEF, 2026–present. Graphene-based 100 Gbps THz communication unit. *Role*: AI/ML lead. Python and C++ pipelines for Transfer Matrix Method optimisation and inverse-problem reconstruction from experimental photonic data; interactive dashboards for spectral and material analysis. Includes a formal industrial secondment to Monozukuri S.p.A. (Rome, IT).
- **THz-Skin** — EU-funded, UEF, 2026–present. THz spectroscopy for accessible skin-cancer detection. *Role*: AI/ML contributor; sample-library and data-analysis workflow design.
- **MEATIC** — Methods for Electromagnetic Analysis of Tokamak Structures, 2026–present. *Role*: data-analysis collaborator; ML workloads on CINECA HPC under project allocation.
- **FUPA1** — University of Cassino. *Role*: data-analysis collaborator; CINECA HPC user under project allocation.
- **CLEPSYDRA** — Interreg Euro-MED. *Role*: co-developer of the end-to-end Python ML and data-ingestion web platform (clepsydra-dss.eu).

AWARDS AND HONOURS

- **Best EvoApps Paper Award**, EvoStar 2026, Toulouse (FR) — conformal prediction in surrogate-assisted evolutionary algorithms [P6].

EDITORIAL ACTIVITY AND SERVICE TO THE SCIENTIFIC COMMUNITY

- **Guest Editor** — Special Issue *AI-Driven Pattern Recognition for Next-Generation Biometrics, Big Data and Cognitive Computing* (MDPI, ISSN 2504-2289). mdpi.com/journal/BDCC/special_issues/7YDN2807HK.
- **Workshop Chair** — BIOMAP @ ICPR 2026 (BIO-inspired Methods for Pattern Recognition).
- **Journal reviewer** — **19 verified reviews for 8 international journals** (reviews in brackets): *Engineering Applications of Artificial Intelligence* (3); *Applied Soft Computing* (3); *Scientific Reports* (3); *Discover Artificial Intelligence* (3); *Intelligence-Based Medicine* (3); *Computers in Human Behavior Reports* (2); *International Journal of Computational Intelligence Systems* (1); *BMC Medical Education* (1).
- **Conference reviewer** — GECCO 2026, PPSN 2026, BIOMAP @ ICPR 2026, AHIA @ ICPR 2026, MCMI @ ICPR 2024.
- **Conference presentations** — EvoStar 2026 Toulouse, 2025 Trieste, 2024 Aberystwyth; ICIAP 2025 Rome; IGS 2025 Montréal and 2023 Évora.

TECHNICAL SKILLS AND CERTIFICATIONS

Programming: Python (expert), C++, C#, Java, SQL; PyTorch, TensorFlow, scikit-learn, pandas, NumPy.

Methods: evolutionary computation, feature selection, multimodal learning, Bayesian networks, conformal prediction, explainable AI (SHAP, attention), agentic AI (LangChain, LangGraph, smolagents).

Infrastructure: Docker, Kubernetes (GPU orchestration), Linux, MongoDB; HPC experience on CINECA Tier-0 (EuroHPC JU).

Certifications: Microsoft Azure AI Fundamentals; Hugging Face AI Agents Fundamentals (2025); AI Literacy — Eduhouse Oy (04/2026, EU AI Act); Master in Europrogettazione (09–12/2024).

Advanced schools: DataSLO Summer School (2025); ACDL 2024 (Deep Learning, Data Science, GenAI); Ethics and AI (2023); Intelligent Sensing Winter School (2022).

THIRD MISSION AND PUBLIC ENGAGEMENT

- **Invited speaker**, Rotaract Club Cassino — podcast series *InnovActionPillsOnAir* on AI and innovation.
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